Category Theory Seminar — Abstracts

University of Coimbra

August 20, 2025

Norms for vectors, linear operators and morphisms of categories, and Cauchy convergence for them

Speaker: Walter Tholen
York University, Canada
Date & Time: May 20, 2025 — 4:00 PM (Coimbra time)

Abstract. In this talk we explore Lawvere's notion of normed category through the lens of easily understood small and large example categories. Cauchy convergence of sequences in such categories is presented as a natural extension of familiar concepts taught in Calculus and Functional Analysis, but has broader impact, also because we allow norms to have values in an arbitrary quantale, rather than just in the real numbers. Our concept fits with the notion of weighted colimit of enriched category theory. There is therefore a natural notion of Cauchy cocompletion of a normed category, the existence proof for which draws on general results from that theory. As an application, Banach's Fixed Point Theorem is extended almost verbatim from metric spaces to normed categories, and it then adds to the large array of results in functorial fixed point theory as studied predominantly by computer scientists.

(Based on joint work with M. M. Clementino and D. Hofmann.)

Self-dual aspects of semi-abelian categories

Speaker: Tim Van der Linden
Université catholique de Louvain and Vrije Universiteit Brussel, Belgium
Date & Time: May 20, 2025 (Tuesday) — 3:00 PM (Coimbra time)
Place: Microsoft Teams

Abstract. The aim of this talk is to introduce di-exact categories [3], which are defined by a simple axiom system capturing self-dual aspects of the context of Janelidze–Marki–Tholen semi-abelian categories [2, 1]. A Borceux–Bourn homological category [1] is Barr-exact if and only if it is di-exact. We explain that classical diagram lemmas such as the Snake Lemma hold in di-exact categories, and give an overview of examples and related contexts.

References

[1] F. Borceux and D. Bourn, *Mal'cev, protomodular, homological and semi-abelian categories*, Math. Appl., vol. 566, Kluwer Acad. Publ., 2004.

[2] G. Janelidze, L. Marki and W. Tholen, Semi-abelian categories, *Journal of Pure and Applied Algebra* **168** (2002), no. 2, 367–386.

[3] G. Peschke and T. Van der Linden, A Homological View of Categorical Algebra, arXiv:2404.15896.

Drazin Inverses in (Dagger) Categories

Speaker: Jean-Simon Pacaud Lemay
Macquarie University, Australia
Date & Time: August 19, 2025 (Tuesday) — 11:00 AM (Coimbra time)
Place: Microsoft Teams

Abstract. Drazin inverses are a special kind of generalized inverse that have been extensively studied and have many applications in ring theory, semigroup theory, and matrix theory. Drazin inverses can also be defined for endomorphisms in any category. A natural question is whether one can extend the notion of Drazin inverse to arbitrary maps—not just endomorphisms. It turns out that this is natural to do in dagger categories. This talk explores Drazin inverses from a categorical perspective, their relation to idempotent splitting, eventual image duality, and introduces *dagger Drazin inverses*. These are connected to Moore–Penrose inverses.

References

- Based on joint work with Robin Cockett and Priyaa Varshinee Srinivasan:
- arXiv:2402.18226
- arXiv:2502.05306

A general approach to pointfree T_0 spaces

Speaker: Anna Laura Suarez
University of Western Cape, South Africa
Date & Time: August 19, 2025 (Tuesday) — 12:00 PM (Coimbra time)
Place: Google Meet

Abstract. The classical dual adjunction between frames and spaces, given by the functors $O\colon \mathbf{Top} \to \mathbf{Frm}$ and $\mathrm{pt}\colon \mathbf{Frm} \to \mathbf{Top}$, restricts to a dual equivalence between sober spaces and spatial frames. Another pointfree approach to the study of spaces is the so-called TD duality, a similar dual adjunction where the fixpoints are the TD spaces. The two dualities are incompatible in that the two spectrum functors are different. Recently, a pointfree approach to spaces whose fixpoints are all T_0 spaces was introduced. The approach is inspired by Raney duality. In this setting, we study the category of Raney extensions and show that we have a dual adjunction extending (O, pt) and capturing all T_0 spaces. By suitably restricting on objects we recover both the classical and the TD duality. Other pointfree notions of T_0 spaces are given by strictly zero-dimensional biframes and McKinsey–Tarski algebras. Aiming to understand the relationship between these three different settings, we introduce and study an abstract notion of pointfree T_0 space. We study the situation where we have a suitable forgetful functor $U\colon \mathcal{C} \to \mathbf{Frm}$ and a dual adjunction $O_{\mathcal{C}}\colon \mathbf{Top} \to \mathcal{C}$ and $\mathrm{pt}_{\mathcal{C}}\colon \mathcal{C} \to \mathbf{Top}$, whose fixpoints are the T_0 spaces and which is, in a sense that will be made precise, compatible with the adjunction (O, pt) between frames and spaces. By studying the fibers of U, we are able to study various aspects of T_0 spaces abstractly. In all pointfree notions of T_0 spaces mentioned above, the pointfree sober spaces are the top elements of the fibers, and the TD ones are the bottom elements.